

## Type of protection Exd – Flameproof enclosure

|          |                                                                                                                                                                                    | Grade of Inspection |       |        |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------|--------|
|          | Check that:                                                                                                                                                                        | Detailed            | Close | Visual |
| <b>A</b> | <b>EQUIPMENT</b>                                                                                                                                                                   |                     |       |        |
| 1        | Equipment is appropriate to the EPL/Zone requirements of the location                                                                                                              | X                   | X     | X      |
| 2        | Equipment group is correct                                                                                                                                                         | X                   | X     |        |
| 3        | Equipment temperature class is correct                                                                                                                                             | X                   | X     |        |
| 4        | Equipment circuit identification is correct                                                                                                                                        | X                   |       |        |
| 5        | Equipment circuit identification is available                                                                                                                                      | X                   | X     | X      |
| 6        | Enclosure, glass parts and glass-to-metal sealing gaskets and/or compounds are satisfactory                                                                                        | X                   | X     | X      |
| 7        | There are no unauthorized modifications                                                                                                                                            | X                   |       |        |
| 8        | There are no visible unauthorised modifications                                                                                                                                    |                     | X     | X      |
| 9        | Bolts, cable entry devices (direct and indirect) and blanking elements are of the correct type and are complete and tight                                                          |                     |       |        |
|          | – physical check                                                                                                                                                                   | X                   | X     |        |
|          | - visual check                                                                                                                                                                     |                     |       | X      |
| 10       | Flange faces are clean and undamaged and gaskets, if any, are satisfactory                                                                                                         | X                   |       |        |
| 11       | Flange gap dimensions are within maximal values permitted                                                                                                                          | X                   | X     |        |
| 12       | Lamp rating, type and position are correct                                                                                                                                         | X                   |       |        |
| 17       | Motor fans have sufficient clearance to enclosure and/or covers                                                                                                                    | X                   |       |        |
| 18       | Breathing and draining devices are satisfactory                                                                                                                                    | X                   | X     |        |
|          |                                                                                                                                                                                    |                     |       |        |
| <b>B</b> | <b>INSTALLATION</b>                                                                                                                                                                |                     |       |        |
| 1        | Type of cable is appropriate                                                                                                                                                       | X                   |       |        |
| 2        | There is no obvious damage to cables                                                                                                                                               | X                   | X     | X      |
| 3        | Sealing of trunking, ducts, pipes and/or conduits is satisfactory                                                                                                                  | X                   | X     | X      |
| 4        | Stopping boxes and cable boxes are correctly filled                                                                                                                                | X                   |       |        |
| 5        | Integrity of conduit system and interface with mixed system is maintained                                                                                                          | X                   |       |        |
| 6        | Earthing connections, including any supplementary earthing bonding connections are satisfactory (for example connections are tight and conductors are of sufficient cross-section) |                     |       |        |
|          | – physical check                                                                                                                                                                   | X                   |       |        |
|          | - visual check                                                                                                                                                                     |                     | X     | X      |
| 7        | Fault loop impedance (TN systems) or earthing resistance (IT systems) is satisfactory                                                                                              | X                   |       |        |
| 8        | Insulation resistance is satisfactory                                                                                                                                              | X                   |       |        |
| 9        | Automatic electrical protective devices operate within permitted limits                                                                                                            | X                   |       |        |
| 10       | Automatic electrical protective devices are set correctly (auto-reset not possible )                                                                                               | X                   |       |        |
| 11       | Specific conditions of use (if applicable) are complied with                                                                                                                       | X                   |       |        |
| 12       | Cables not in use are correctly terminated                                                                                                                                         | X                   |       |        |
| 13       | Obstructions adjacent to flameproof flanged joints are in accordance with IEC 60079-14                                                                                             | X                   | X     | X      |
| 14       | Variable voltage/frequency installation in accordance with documentation                                                                                                           | X                   | X     |        |
| <b>C</b> | <b>ENVIRONMENT</b>                                                                                                                                                                 |                     |       |        |

|   |                                                                                                   |   |   |   |
|---|---------------------------------------------------------------------------------------------------|---|---|---|
| 1 | Equipment is adequately protected against corrosion, weather, vibration and other adverse factors | X | X | X |
| 2 | No undue accumulation of dust and dirt                                                            | X | X | X |
| 3 | Electrical insulation is clean and dry                                                            |   |   |   |

## Ex e

### Grade of Inspection

|          | Check that:                                                                                                                                                                        | Detailed | Close | Visual |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|--------|
| <b>A</b> | <b>EQUIPMENT</b>                                                                                                                                                                   |          |       |        |
| 1        | Equipment is appropriate to the EPL/Zone requirements of the location                                                                                                              | X        | X     | X      |
| 2        | Equipment group is correct                                                                                                                                                         | X        | X     |        |
| 3        | Equipment temperature class is correct                                                                                                                                             | X        | X     |        |
| 4        | Equipment circuit identification is correct                                                                                                                                        | X        |       |        |
| 5        | Equipment circuit identification is available                                                                                                                                      | X        | X     | X      |
| 6        | Enclosure, glass parts and glass-to-metal sealing gaskets and/or compounds are satisfactory                                                                                        | X        | X     | X      |
| 7        | There are no unauthorized modifications                                                                                                                                            | X        |       |        |
| 8        | There are no visible unauthorised modifications                                                                                                                                    |          | X     | X      |
| 9        | Bolts, cable entry devices (direct and indirect) and blanking elements are of the correct type and are complete and tight                                                          |          |       |        |
|          | – physical check                                                                                                                                                                   | X        | X     |        |
|          | - visual check                                                                                                                                                                     |          |       | X      |
| 10       | Flange faces are clean and undamaged and gaskets, if any, are satisfactory                                                                                                         |          |       |        |
| 11       | Flange gap dimensions are within maximal values permitted                                                                                                                          |          |       |        |
| 12       | Lamp rating, type and position are correct                                                                                                                                         | X        |       |        |
| 13       | Electrical connections are tight                                                                                                                                                   | X        |       |        |
| 14       | Condition of enclosure gaskets is satisfactory                                                                                                                                     | X        |       |        |
| 17       | Motor fans have sufficient clearance to enclosure and/or covers                                                                                                                    | X        |       |        |
| 18       | Breathing and draining devices are satisfactory                                                                                                                                    | X        | X     |        |
|          |                                                                                                                                                                                    |          |       |        |
| <b>B</b> | <b>INSTALLATION</b>                                                                                                                                                                |          |       |        |
| 1        | Type of cable is appropriate                                                                                                                                                       | X        |       |        |
| 2        | There is no obvious damage to cables                                                                                                                                               | X        | X     | X      |
| 3        | Sealing of trunking, ducts, pipes and/or conduits is satisfactory                                                                                                                  | X        | X     | X      |
| 5        | Integrity of conduit system and interface with mixed system is maintained                                                                                                          | X        |       |        |
| 6        | Earthing connections, including any supplementary earthing bonding connections are satisfactory (for example connections are tight and conductors are of sufficient cross-section) |          |       |        |
|          | – physical check                                                                                                                                                                   | X        |       |        |
|          | - visual check                                                                                                                                                                     |          | X     | X      |
| 7        | Fault loop impedance (TN systems) or earthing resistance (IT systems) is satisfactory                                                                                              | X        |       |        |
| 8        | Insulation resistance is satisfactory                                                                                                                                              | X        |       |        |
| 9        | Automatic electrical protective devices operate within permitted limits                                                                                                            | X        |       |        |
| 10       | Automatic electrical protective devices are set correctly (auto-reset not possible )                                                                                               | X        |       |        |
| 11       | Specific conditions of use (if applicable) are complied with                                                                                                                       | X        |       |        |
| 12       | Cables not in use are correctly terminated                                                                                                                                         | X        |       |        |
|          | IEC 60079-14                                                                                                                                                                       |          |       |        |

|          |                                                                                                   |   |   |   |
|----------|---------------------------------------------------------------------------------------------------|---|---|---|
| 14       | Variable voltage/frequency installation in accordance with documentation                          | X | X |   |
| <b>C</b> | <b>ENVIRONMENT</b>                                                                                |   |   |   |
| 1        | Equipment is adequately protected against corrosion, weather, vibration and other adverse factors | X | X | X |
| 2        | No undue accumulation of dust and dirt                                                            | X | X | X |
| 3        | Electrical insulation is clean and dry                                                            | X |   |   |

## Ex n

### Grade of Inspection

|          | Check that:                                                                                                                                                                        | Detailed | Close | Visual |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|--------|
| <b>A</b> | <b>EQUIPMENT</b>                                                                                                                                                                   |          |       |        |
| 1        | Equipment is appropriate to the EPL/Zone requirements of the location                                                                                                              | X        | X     | X      |
| 2        | Equipment group is correct                                                                                                                                                         | X        | X     |        |
| 3        | Equipment temperature class is correct                                                                                                                                             | X        | X     |        |
| 4        | Equipment circuit identification is correct                                                                                                                                        | X        |       |        |
| 5        | Equipment circuit identification is available                                                                                                                                      | X        | X     | X      |
| 6        | Enclosure, glass parts and glass-to-metal sealing gaskets and/or compounds are satisfactory                                                                                        | X        | X     | X      |
| 7        | There are no unauthorized modifications                                                                                                                                            | X        |       |        |
| 8        | There are no visible unauthorised modifications                                                                                                                                    |          | X     | X      |
| 9        | Bolts, cable entry devices (direct and indirect) and blanking elements are of the correct type and are complete and tight                                                          |          |       |        |
|          | – physical check                                                                                                                                                                   | X        | X     |        |
|          | - visual check                                                                                                                                                                     |          |       | X      |
| 12       | Lamp rating, type and position are correct                                                                                                                                         | X        |       |        |
| 13       | Electrical connections are tight                                                                                                                                                   | X        |       |        |
| 14       | Condition of enclosure gaskets is satisfactory                                                                                                                                     | X        |       |        |
| 15       | Enclosed-break and hermetically sealed devices are undamaged                                                                                                                       | X        |       |        |
| 16       | Restricted breathing enclosure is satisfactory                                                                                                                                     | X        |       |        |
| 17       | Motor fans have sufficient clearance to enclosure and/or covers                                                                                                                    | X        |       |        |
| 18       | Breathing and draining devices are satisfactory                                                                                                                                    | X        | X     |        |
| <b>B</b> | <b>INSTALLATION</b>                                                                                                                                                                |          |       |        |
| 1        | Type of cable is appropriate                                                                                                                                                       | X        |       |        |
| 2        | There is no obvious damage to cables                                                                                                                                               | X        | X     | X      |
| 3        | Sealing of trunking, ducts, pipes and/or conduits is satisfactory                                                                                                                  | X        | X     | X      |
| 5        | Integrity of conduit system and interface with mixed system is maintained                                                                                                          | X        |       |        |
| 6        | Earthing connections, including any supplementary earthing bonding connections are satisfactory (for example connections are tight and conductors are of sufficient cross-section) |          |       |        |
|          | – physical check                                                                                                                                                                   | X        |       |        |
|          | - visual check                                                                                                                                                                     |          | X     | X      |
| 7        | Fault loop impedance (TN systems) or earthing resistance (IT systems) is satisfactory                                                                                              | X        |       |        |
| 8        | Insulation resistance is satisfactory                                                                                                                                              | X        |       |        |
| 9        | Automatic electrical protective devices operate within permitted limits                                                                                                            | X        |       |        |
| 10       | Automatic electrical protective devices are set correctly (auto-reset not possible )                                                                                               | X        |       |        |
| 11       | Specific conditions of use (if applicable) are complied with                                                                                                                       | X        |       |        |
| 12       | Cables not in use are correctly terminated                                                                                                                                         | X        |       |        |
|          | IEC 60079-14                                                                                                                                                                       |          |       |        |

|          |                                                                                                   |   |   |   |
|----------|---------------------------------------------------------------------------------------------------|---|---|---|
| 14       | Variable voltage/frequency installation in accordance with documentation                          | X | X |   |
| <b>C</b> | <b>ENVIRONMENT</b>                                                                                |   |   |   |
| 1        | Equipment is adequately protected against corrosion, weather, vibration and other adverse factors | X | X | X |
| 2        | No undue accumulation of dust and dirt                                                            | X | X | X |
| 3        | Electrical insulation is clean and dry                                                            | X |   |   |

## Ex tD

### Grade of Inspection

|          | Check that:                                                                                                               | Detailed | Close | Visual |
|----------|---------------------------------------------------------------------------------------------------------------------------|----------|-------|--------|
| <b>A</b> | <b>EQUIPMENT</b>                                                                                                          |          |       |        |
| 1        | Equipment is appropriate to the EPL/Zone requirements of the location                                                     | X        | X     | X      |
| 2        | IP grade of equipment is appropriate to conductivity of dust                                                              | X        | X     | X      |
| 3        | Equipment maximum surface temperature is correct                                                                          | X        | X     |        |
| 4        | Equipment circuit identification is correct                                                                               | X        | X     | X      |
| 5        | Equipment circuit identification is available                                                                             | X        |       |        |
| 6        | Enclosure, glass parts and glass-to-metal sealing gaskets and/or compounds are satisfactory                               | X        | X     | X      |
| 7        | There are no unauthorized modifications                                                                                   | X        |       |        |
| 8        | There are no visible unauthorised modifications                                                                           |          | X     | X      |
| 9        | Bolts, cable entry devices (direct and indirect) and blanking elements are of the correct type and are complete and tight |          |       |        |
|          | – physical check                                                                                                          | X        | X     |        |
|          | - visual check                                                                                                            |          |       | X      |
| 10       | Lamp rating, type and position are correct                                                                                | X        |       |        |
| 11       | Electrical connections are tight                                                                                          | X        |       |        |
| 12       | Condition of enclosure gaskets is satisfactory                                                                            | X        |       |        |
| 13       | Motor fans have sufficient clearance to enclosure and/or covers                                                           | X        |       |        |
| <b>B</b> | <b>INSTALLATION</b>                                                                                                       |          |       |        |
| 1        | The installation is such as to minimize the risk of dust accumulations                                                    | X        | X     | X      |
| 2        | Type of cable is appropriate                                                                                              | X        |       |        |
| 3        | There is no obvious damage to cables                                                                                      | X        | X     | X      |
| 4        | Sealing of trunking, ducts, pipes and/or conduits is satisfactory                                                         | X        | X     | X      |
| 5        | Earthing connections, including any supplementary earthing bonding connections are satisfactory                           |          |       |        |
|          | – physical check                                                                                                          | X        |       |        |
|          | - visual check                                                                                                            |          | X     | X      |
| 6        | Fault loop impedance (TN systems) or earthing resistance (IT systems) is satisfactory                                     | X        |       |        |
| 7        | Insulation resistance is satisfactory                                                                                     | X        |       |        |
| 8        | Automatic electrical protective devices operate within permitted limits                                                   | X        |       |        |
| 9        | Automatic electrical protective devices are set correctly (auto-reset not possible )                                      | X        |       |        |
| 10       | Specific conditions of use (if applicable) are complied with                                                              | X        |       |        |
| 11       | Cables not in use are correctly terminated                                                                                | X        | X     |        |
|          | IEC 60079-14                                                                                                              |          |       |        |
| <b>C</b> | <b>ENVIRONMENT</b>                                                                                                        |          |       |        |
| 1        | Equipment is adequately protected against corrosion, weather, vibration and other adverse factors                         | X        | X     | X      |

|   |                                        |   |   |   |
|---|----------------------------------------|---|---|---|
| 2 | No undue accumulation of dust and dirt | X | X | X |
|---|----------------------------------------|---|---|---|

## 12e 'p' 'pD'

(IEC 60079-2 for gases or IEC 61241-4 for dusts.)

### Grade of Inspection

|          | Check that:                                                                                                   | Detailed | Close | Visual |
|----------|---------------------------------------------------------------------------------------------------------------|----------|-------|--------|
| <b>A</b> | <b>EQUIPMENT</b>                                                                                              |          |       |        |
| 1        | Equipment is appropriate to the EPL/Zone requirements of the location                                         | X        | X     | X      |
| 2        | Equipment group is correct                                                                                    | X        | X     |        |
| 3        | Equipment temperature class or surface temperature is correct                                                 | X        | X     |        |
| 4        | Equipment circuit identification is correct                                                                   | X        |       |        |
| 5        | Equipment circuit identification is available                                                                 | X        | X     | X      |
| 6        | Enclosure, glass parts and glass-to-metal sealing gaskets and/or compounds are satisfactory                   | X        | X     | X      |
| 7        | There are no unauthorized modifications                                                                       | X        |       |        |
| 8        | There are no visible unauthorised modifications                                                               |          | X     | X      |
| 9        | Lamp rating, type and position are correct                                                                    | X        |       |        |
|          |                                                                                                               |          |       |        |
| <b>B</b> | <b>INSTALLATION</b>                                                                                           |          |       |        |
| 1        | Type of cable is appropriate                                                                                  | X        |       |        |
| 2        | There is no obvious damage to cables                                                                          | X        | X     | X      |
| 3        | Earthing connections, including any supplementary earthing bonding connections are satisfactory               |          |       |        |
|          | – physical check                                                                                              | X        |       |        |
|          | - visual check                                                                                                |          | X     | X      |
| 4        | Fault loop impedance (TN systems) or earthing resistance (IT systems) is satisfactory                         | X        |       |        |
| 5        | Automatic electrical protective devices operate within permitted limits                                       | X        |       |        |
| 6        | Automatic electrical protective devices are set correctly                                                     | X        |       |        |
| 7        | Protective gas inlet temperature is below maximum specified                                                   | X        |       |        |
| 8        | Ducts, pipes and enclosures are in good condition                                                             | X        | X     | X      |
| 9        | Protective gas is substantially free from contaminants                                                        | X        | X     | X      |
| 10       | Protective gas pressure and/or flow is adequate                                                               | X        |       |        |
| 11       | Pressure and/or flow indicators, alarms and interlocks function correctly                                     | X        |       |        |
| 12       | Conditions of spark and particle barriers of ducts for exhausting the gas in hazardous areas are satisfactory | X        |       |        |
| 13       | Specific conditions of use (if applicable) are complied with                                                  | X        |       |        |
|          |                                                                                                               |          |       |        |
| <b>C</b> | <b>ENVIRONMENT</b>                                                                                            |          |       |        |
| 1        | Equipment is adequately protected against corrosion, weather, vibration and other adverse factors             | X        | X     | X      |
| 2        | No undue accumulation of dust and dirt                                                                        | X        | X     | X      |